higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 2

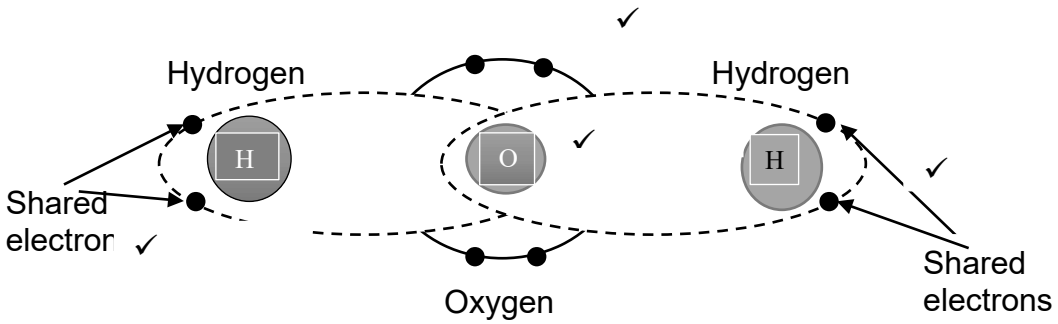
1 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1: COMPONENTS AND CIRCUIT DRAWING

1.1



(4)

1.2 1.2.1 Numerical marking
1.2.2 Explicit marking

(2 × 1) (2)

1.3 1.3.1 470 pF (1)
1.3.2 10 µF 16 V (1)

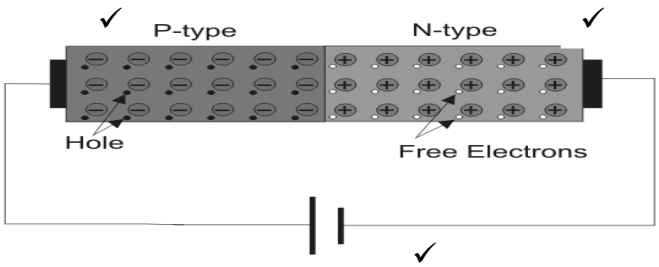
1.4 1.4.1 A capacitor is an electronic component used to store an electrical charge.

1.4.2 A transformer is a device that transfers electrical energy from one circuit to another by electromagnetic induction.

1.4.3 A relay is an electrically powered switch.

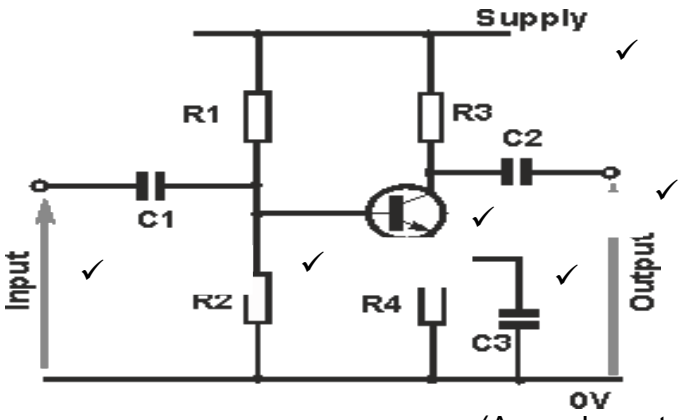
(3 × 1) (3)

1.5



(3)

1.6



(Any relevant correct sketch) (6)

[20]

QUESTION 2: COMPONENTS AND CIRCUIT DRAWING

2.1 The inputs and outputs of linear integrated circuits can take on a continuous range of values and the outputs are generally proportional to the inputs. Examples: Operational amplifier, small signal amplifiers, audio power amplifiers and voltage regulators (2)

2.2 Pin 6 is for the output voltage. (1)

2.3 $V_{in} = 2\ V$ $R_f = 30\ k\Omega$ $R_1 = 2\ \Omega$

$$V_o = \left[1 + \frac{R_f}{R_1}\right] VIN$$

$$\therefore \left(1 + \frac{3\ 000\ \Omega}{2\ 000\ \Omega}\right)(2\ V) \checkmark$$

$$= 5\ V \checkmark$$ (3)

2.4
A Centre-tapped transformer
B Rectifier diode
C Zener diode
D Resistor
E Polarised capacitor
F NPN-transistor

(6 × 1) (6)

2.5
- Connect the red lead (+) to the anode and black lead (-) to the cathode.
- The diode will conduct and the meter will display a value.
- Reverse the connection and connect the red lead (+) to the cathode and the black lead (-) to the anode.
- The diode should not conduct. (4)

2.6
- It is easy to change connections.
- Parts cannot be damaged as no soldering is used.
- It is easy to change components and note change in the behaviour of the circuit. (3)

2.7
- Avoid inhaling smoke by keeping your head to the side and not above the work.
- Minimise smoke and fumes by keeping the soldering iron clean – wipe it on a damp sponge before putting it back in its stand.
- Avoid blowing a fan directly on the work.
- Always wash your hands after handling solder wire since the lead in solder is extremely toxic. (Any relevant 3 × 1) (3)

2.8 $L_1 = 3\ mH$ $L_2 = 5\ mH$

$$L_T = \frac{L_1 \times L_2}{L_2 + L_2}$$

$$\therefore \left(\frac{3 \times 5}{3 + 5}\right) \times 10^{-3}$$ (2)

$= 1{,}875 \ \ mH$ ✓

2.9 The maximum current is 20 mA. (1)
[25]

QUESTION 3: DIGITAL ELECTRONICS

3.1
- They are less affected by noise.
- They are easier to design.
- They have more flexibility and functionality.
- They operate faster.
- They are more reliable.
- They store digital data easier.
- They are more compactible and use less space. (Any 4 × 1) (4)

3.2
- 1 is used to represent the higher voltage level – HIGH
- 0 is used to represent the lower voltage level – LOW (2)

3.3 An analogue signal is continuously variable✓ while a digital signal is a representation of a physical quantity by a series of binary numbers.✓ (2)

3.4
$$\begin{array}{cccccc} 1 & 0 & 1 & 1 & 0 & 1 \\ 32 & 16 & 8 & 4 & 2 & 1 \end{array}$$
¿$32 + 8 + 4 + 1$✓
$= 45_{10}$✓ (Any relevant 4 × 1) (4)

3.5
$\frac{35}{2} = 17{,}5 \quad rem \quad 1$
$\frac{17}{2} = 8{,}5 \quad rem \quad 1$
$\frac{8}{2} = 4 \quad rem \quad 0$ ✓
$\frac{4}{2} = 2 \quad rem \quad 0$
$\frac{2}{2} = 1 \quad rem \quad 0$
$\frac{1}{2} = 0{,}5 \quad rem \quad 1$
$= 100011_2$

$\frac{14}{2} = 7 \quad rem \quad 0$
$\frac{7}{2} = 3{,}5 \quad rem \quad 1$✓
$\frac{3}{2} = 1{,}5 \quad rem \quad 1$
$\frac{1}{2} = 0{,}5 \quad rem \quad 1$
$= 1110_2$

$$\begin{array}{r} 100011 \\ +\ \ 1110 \\ \hline \end{array}$$
¿ 110001 (Any relevant 8 × 1) (8)

3.6
$$\begin{array}{rr} 1011 & +\ 1011 \\ -\ \ 1100 = \text{1s comp} & 0011 \\ \hline \text{¿}\ \ 1111 & 1110 \text{ no carry – recomplement } 0001 \end{array}$$
(2)

3.7

A (input)	B (input)	X (output)
0	0	0✓
0	1	0✓
1	0	0✓
1	1	1✓

An AND gate produces only a 1 output when all the inputs are 1.✓✓ (4)

3.8 3.8.1



3.8.2



3.8.3



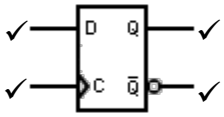
3.8.4



(4 × 1) (4)

3.9 A flip-flop is a circuit with two stable states that can be used to store information. (1)

3.10



(4)
[35]

QUESTION 4: BASIC COMPUTER COMPONENTS

4.1 The main function of computer ports is to serve as an interface between the computer and other components. (2)

4.2 4.2.1 A monitor is used to display data or information for the user.

4.2.2 A printer is used to print a paper copy of what one sees on the monitor or what is stored in the computer.

4.2.3 A scanner is used to transfer pictures and photographs to a computer.

(3 × 1) (3)

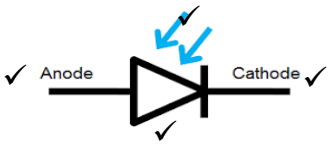
4.3 A CD is read by focusing a laser through the bottom of the polycarbonate layer.✓ The change in height between pits and lands leads to a difference in the amount of light reflected.✓ A light sensor detects a change in reflection when it encounters a change from pit to land or land to pit.✓ Each change is a 1.✓ The lack of change is 0 s.✓ (Any relevant answer) (5)
[10]

QUESTION 5: TRANSDUCER

5.1
- Conductive probes
- Capacitive probes
- Ultrasonic probes

(Any 2 × 1) (2)

5.2



Photodiode symbol (4)

5.3
- They are simple to operate.
- They are very reliable. (2)

5.4
- Concentric orifice plates
- Segmental orifice plates
- Eccentric orifice plates

(Any 2 × 1) (2)
[10]

TOTAL: **100**

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